Contents

One World, One Cut: A Real-Wave Interpretation of Quantum Measurement after a Failed Mechanism

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*[DRAFT v1.0 — 2026-09-04. The single paper into which the program's three papers collapse, at the sponsor's decision of the same day ("the three papers collapse down to one"), written as the rewrite of Paper 3 without synchronization. Provenance, kept whole: Paper 1 (*A Field–Matter Selector for Outcome Production*, v0.9) is now the program's recorded negative result (`NEGATIVE_RESULT.md`; its ledger and race package remain as the record); Paper 2 (*The Heisenberg Cut as a Physical Threshold*, v0.4) is absorbed here as §3 and §6, with its test program (`heisenberg_cut_recoverability/`) as the evidence; Paper 3 v0.1 (*One Mechanism at Two Scales: The Dirac–Kuramoto Framework*) is superseded and stays in the repository as the record of the synchronization program, as the Many Clocks Interpretation stayed before it. Everything here that depended on phase locking as the mechanism of selection has been removed or demoted to amplifier phenomenology; everything that survived its tests is kept and cited to the test. Byline order per standing convention, the sponsor's decision before any circulation.]*

[v1.1 — 2026-09-05. Claim-level changes propagated from the publication candidate's revision after the four-reviewer internal round (`one_world_one_cut_AI_review_2026-09/`), so that this record and the candidate do not diverge; each is marked "v1.1" in place. (1) The selection postulate's clause 2 is corrected: a hazard on the whole quantum linear in local drawn energy is falsified by the record's own race once channels are reached at different times (`heisenberg_cut_recoverability/REVIEW_RUNS_RESULTS.md`, Run B); the hazard acts on the branch weight of the conditional state — the quantum-jump unravelling — and the no-jump renormalization is nonlocal. (2) The cut's location has a regime: coupling-set (half-width 2K) for $\Gamma \ll K$, record-set ($\Gamma/2$) and hence temperature-set for $\Gamma \gg K$, the regime of room-temperature semiconductor detectors (Run A); "coupling-set rather than temperature-set" is withdrawn as a distinctive claim. (3) The in-principle recoverability claim of §3.5 is conceded to decoherence theory. (4) No-signalling is a theorem of the Lindblad average of the unravelling, not a consequence of hazard linearity. (5) The frame tests are withdrawn as tests of the interpretation; the foliation is a choice with relativistic-collapse alternatives. (6) The paper's one result is a no-go. The short form carries the full rewrite and the corrected citations; this long form keeps its development history and its numbers as the record.]

Contributions. Claude Fable 5 (Anthropic) performed the formalization, the manuscript prose, the simulations and their records, and the literature work. John M. Bramble, MD supplied the physical framing and research direction — the real-wave commitment, the register and clock readings, the time–energy–information binding, the temperature dial, and the decision to record the failed mechanism rather than bury it — adjudicated scope and interpretation, and is the accountable human sponsor.

Abstract. We state an interpretation of quantum measurement and price it. Its commitments are four: the wave is real (the Dirac field for matter, the Maxwell field for light); there is one world; the Heisenberg cut is a physical crossover with a computable location, a computable width, and a physical address, the absorption vertex of a detector site; and outcome selection is a stochastic event at that vertex, with the hazard of the golden rule, put in as a postulate. The interpretation was reached by subtraction. A three-paper program set out to derive the last of these — the Born weights and the one-outcome property — from a synchronization dynamics in a real-wave substrate, and failed: the mechanism produced the Born square only by containing the golden rule's structure and a nonlocal one-quantum constraint, neither of which it derived. The failure is recorded, and what survived it is this paper. The cut's location, κ_ret/K = 1 with κ_ret the deficit return rate and K the coupling's population-transfer rate, passed its first exact test; its width, a layer of share K/ω, is a crossover of relative width order one in the deficit-rate variable, as measured; a dense record channel is a physical requirement; the carrier frequency enters only as a Bloch–Siegert shift; a self-sustaining amplifier kinks the recoverability crossover rather than sharpening it. All of this is compatible with decoherence theory, and the paper says so. What it adds is narrow and is stated as such (v1.1): a location for the event with its regime — coupling-set where the coupling exceeds the record channel's rate, record-set and hence temperature-set where it does not; the dense-record requirement; the distinction between in-principle recoverability and the reversal an experimenter can apply; and a negative result — that a real-wave, one-world programme with no new constant, forced to reproduce quantum mechanics, is pushed to the projection postulate's branch weights at the decoherence crossover, and that nothing in the physics of a detector or in a synchronization substrate can move the outcome weights. The v1.0 claims that recoverability ends in principle and that the crossing is coupling-set rather than temperature-set are withdrawn. The nearest existing interpretation is Cramer's transactional one, and the relation is drawn exactly. No new constant is introduced and no prediction differs from quantum mechanics; the paper's experimental content is a set of consistency tests and one falsifiable premise.

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1. What this paper is, and what it is not

It is an interpretation. It makes no prediction that differs from quantum mechanics as ordinarily used, introduces no constant, and modifies no equation. What it offers is an account of what the equations describe and of where, physically, the description stops being reversible. Its four commitments — real waves, one world, a located cut, selection at absorption — are stated in §2, §3, and §4, and the price of each is stated where it is incurred.

It is the residue of a program that failed. Between 2026-07 and 2026-09 this program pursued a stronger claim: that a real-wave substrate of phase-locked Dirac clocks could produce the Born statistics and the one-outcome property from dynamics, without the square being put in by hand. That claim was tested by simulation at every joint a ledger could name, with the predictions written before the results were opened, and it did not survive. The record is `NEGATIVE_RESULT.md` and the ledger and race package behind it; §4.2 states what was learned. The present paper keeps what the tests left standing and nothing else. Its title records the method: an interpretation reached by subtraction.

Three things not to conflate. Measurement talk chronically merges three processes, and the interpretive disputes are disputes about the third. (1) *Entangling interaction*: couple system to apparatus; on the joint Hilbert space this is unitary evolution, linear and reversible, and nothing here distinguishes a measurement from any other interaction, which is von Neumann's point and the origin of the chain. (2) *Decoherence*: open the apparatus to an environment and trace it out; the reduced state obeys an irreversible master equation that is still linear in the density matrix, so off-diagonal terms are suppressed at spectacular rates and the state comes to look like a mixture — but a linear map sends a superposition to a mixture of its branches, weighted exactly as before, never to one of them. Irreversibility has been purchased; selection has not. (3) *The event*: one outcome, in one world. Everett denies it exists; GRW/CSL postulate it as new stochastic dynamics with new constants; Penrose supplies a scale without a mechanism; decoherence-only accounts stop at (2) and redescribe the question. This paper affirms that (3) occurs, places it at the absorption vertex of a detector site, and — this is the change from the reviewed drafts — postulates it rather than deriving it.

Lineage. The cut is the decoherence programme's home ground, and the relation must be stated exactly. Zeh, Zurek, and Joos and Zeh established that the placement of the split is dynamical, not arbitrary: the environment monitors the apparatus in a pointer basis it selects, and interference between pointer states becomes unobservable at a rate the coupling sets [Zeh 1970; Zurek 1981, 1982; Joos & Zeh 1985]. Tegmark made the consequence explicit as apparent collapse caused by scattering [Tegmark 1993]; Brune and colleagues watched a cavity-field meter decohere progressively as its states were made more distinguishable, a measured crossover of the kind §3 describes [Brune et al. 1996]; quantum Darwinism turned the environment's record into the criterion of objectivity [Ollivier et al. 2004; Zurek 2009]. Schlosshauer's review states the programme's own limit, that decoherence explains the appearance of a cut and not the occurrence of one outcome [Schlosshauer 2004]; and Schlosshauer and Camilleri, asking whether decoherence challenges Bohr's doctrine that a classical side must exist, conclude that it justifies where the cut may be placed without making the cut a physical discontinuity [Schlosshauer & Camilleri 2008; Camilleri & Schlosshauer 2015] — the "for all practical purposes" that Bell would not accept [Bell 1990]. Everything in §3 is compatible with that lineage. What this paper adds to it is in §3.5 and §4, and nowhere else.

2. The ontology

2.1 Real waves

The wave is a physical field: the Dirac field for matter, the Maxwell field for light, on ordinary spacetime for one excitation and on configuration space for several (the second of Schrödinger's two defeats, held open in §5, not glossed). "Particle" names a persisting excitation; persistence is a property of bound states — the displaced pole of a dressed excitation, off its free shell and held there by its binding — and needs no mechanism beyond the equations. The reviewed drafts made persistence a lock; this paper makes it what the standard theory already makes it, and loses nothing by the retreat.

The representation-independent statement of the virtual/real distinction is *propagating versus evanescent*: real wavenumber, phase advancing, energy carried away; versus imaginary wavenumber, exponentially damped, clinging to its source. Frustrated total internal reflection is the evanescent excitation made tangible. No one disputes that this boundary is physical, and it is the only virtual/real boundary this paper uses.

2.2 Superposition as coexistence, and the register

A superposition is both components present at once with a definite relative phase — coexistence, never alternation. When the components differ in energy their beat at the Bohr frequency $\Delta E/\hbar$ is the laboratory signature that both are there: quantum beats, Ramsey fringes, and the founding instance, Schrödinger's *zitterbewegung*, the interference of the positive- and negative-energy components of a Dirac wavepacket at $2mc^2/\hbar$ [Schrödinger 1930]. Degenerate superpositions are the silent case, both components present with a frozen relative phase and no beat.

The Born-compatibility constraint is geometric. On the Bloch sphere the free dynamics precesses the phase at fixed latitude; the outcome populations are constants of the unitary motion. Literal alternation between outcomes would time-average every superposition to even odds and destroy the Born weights. The latitude carries the weight; the phase is what evolves; and the move from latitude to pole is the event of §4, which is not unitary.

A single two-sector structure recurs: the electron's chiral pair, the photon's helicity pair, the interferometer's which-arm pair — an SU(2) register storing its observable as a relative amplitude and phase, intrinsic in the first two cases and geometric in the third, where a beamsplitter sculpts the mode function into two branches. Path superposition is therefore not a puzzle about a delocalized photon; it *is* the delocalization, shaped — a mode function with two branches is no stranger than a wide packet, and the single-photon anticorrelation-plus-interference experiment [Grangier et al. 1986] shows both in one apparatus. Polarization is the helicity register: the Poincaré sphere is its Bloch sphere, and rotating the R–L phase by φ rotates the linear axis by $\varphi/2$, the half-angle that makes polarization behave spinorially at spin 1. The clean division the register enforces: **delocalization is the wave's birthright; discreteness is the detector's budget** — one quantum, one closure, however thinly the amplitude is spread. Which-path is the register electromagnetic detectors einselect, because the interaction couples through the vector current; it is the superposition detectors are best at destroying.

2.3 The photon as the uncoupled limit

Vacuum Maxwell in Riemann–Silberstein form, $\mathbf{F}\pm = \mathbf{E} \pm i\mathbf{B}$, splits into two decoupled Weyl-form equations, the two helicity sectors [Bialynicki-Birula 1996]. The photon is the register with the off-diagonal coupling set to zero: no mass term, no gap, no beat, no internal clock. The condensate that gives every fermion its mass — the Yukawa coupling is by definition a chirality flip, so $m = y\langle\varphi\rangle$ — leaves the photon massless because the residual U(1) is unbroken. Matter can lend the photon a coupling (a plasma an effective mass; a magnetized or chiral medium a helicity splitting), so everything clock-like about light is borrowed. Two caveats fence the correspondence: the sectors are spin-1 objects, not spinors; and $|\mathbf{F}|^2$ is an energy density, not a Born amplitude.

2.4 The clock, sober

A massive excitation carries a phase advancing at the Compton frequency $mc^2/\hbar$ in its rest frame. This is de Broglie's clock, and it is not a metaphor: an atom interferometer whose fringes are set directly by the caesium mass-frequency has been built [Lan et al. 2013]. The Dirac phase runs at $\omega\_C = mc^2/\hbar$ while the observable beat of a wavepacket containing both energy signs runs at $2\omega\_C$ — a spinor returns to itself after 720°, every bilinear after 360°. Two sobrieties the reviewed drafts lacked. The chiral oscillation is a feature of the Dirac representation, not an invariant: the Foldy–Wouthuysen transformation removes it for a positive-energy state [Foldy & Wouthuysen 1950], and in field theory the mass term is the amplitude for a chirality flip, a superposition of two chiralities rather than a hand moving on a dial. And a phase is not something a classical oscillator can catch: its frequency is 10^{21} Hz against anything in a detector, the linear evolution that carries it has no autonomous phase to entrain, and the photon has none at all. The clock is real; it is not a coupling. What the framework once called "Dirac supplies the clock, Kuramoto supplies the lock" now reads: Dirac supplies the clock, and nothing locks it.

2.5 Two classical limits

Matter becomes classical by being *recorded*: one absorption at a time, dot by dot, the electron interference pattern built up from single events [Tonomura et al. 1989]; there is no classical electron wave, because Pauli exclusion bars a fermion field from macroscopic occupation of one mode. Light becomes classical by *occupancy*: the macroscopically occupied single mode with a common phase, the coherent state [Glauber 1963], loud rather than averaged, fluctuations $\sqrt{N}$ against signal N — a route Bose statistics permits and stimulated emission enhances. Thermal light is classical only as noise, intensities and bunching [Hanbury Brown & Twiss 1956] with no definite phase; a large-N number state is loud and phaseless; the missing common phase is, in the laser, manufactured on the matter side by a threshold transition in the gain medium [Haken 1975]. Matter reaches the occupancy route only by first becoming bosonic (Cooper pairs, condensates). Spin-statistics distributes the two limits; each species takes the only road open to it. A single photon, too free to be recorded in flight and too small to be loud, is excluded from both: in flight it has no lock partner (stellar interferometry survives light-years), and at interaction it is annihilated, converting itself into a record in matter. None of this is a new empirical claim; it is the organizing pattern a century of experiments already follows, from Young's manufactured coherence and Einstein's two-term fluctuation formula through Taylor's feeble light and Hanbury Brown–Twiss to the laser and Tonomura, and the framework names the axes the century was measuring along without claiming any of it as evidence for itself over standard quantum mechanics.

2.6 One world

Exactly one outcome occurs. This is a commitment, and Bell's theorem presents its bill at once: a single world with definite outcomes that reproduces the quantum correlations is nonlocal. Everett escapes the bill by denying single outcomes, buying locality with many worlds. This paper makes the opposite trade and presents it as a trade, not a conquest. What it pays is stated in §5.

3. The cut, with an address

This section is the surviving content of the program's second paper, condensed, with each claim cited to the test that bears on it. The tests are exact calculations in the smallest models that contain the relevant physics, with predictions fixed before running; their records, including the predictions that were wrong, are in `heisenberg_cut_recoverability/`.

3.1 The criterion

The cut is where **in-principle recoverability ends**: the first point in the chain at which no operation, however idealized, can restore the pre-event configuration. Operationally, recoverability is the partial Loschmidt echo — the site's own Hamiltonian reversed for the capture time, the environment left untouched. It is not the reversal an experimenter can apply to the site–field coupling alone: even with no environment at all, flipping the coupling returns a detuned capture only on resonance, because the detuning keeps running forward and the flipped coupling tilts the rotation the other way instead of undoing it (the coupling-flip return fell to 0.04 off resonance while the echo returned the quantum to one part in 10^{15}). Any experiment that locates the cut by a reversal protocol must separate the two quantities.

3.2 The location

Below the shell, a site holding $e < E_0$ is off-shell with deficit $\Delta E = E_0 - e$, and the deficit is a return rate, κ_ret = ΔE/ħ: the rate at which the virtual excitation returns to the common field. This is not bath dissipation; the returned energy re-enters the field. The cut sits where the return rate balances the coupling: **κ_ret/K = 1**, with K the rate at which the coupling moves population.

First exact test. The smallest model that spans both sides with one continuous dynamics — one photon mode, one absorber detuned by Δ from the shell so that its off-shell excitation returns at κ_ret = Δ, and a dense record channel of 256 modes leaking at a golden-rule rate Γ — shows a crossover, not a bifurcation. Recoverability leaks at Γ times the fraction of the time the excitation spends on the site; that fraction is $2K^2/(\Delta^2 + 4K^2)$, with half-point $\Delta = 2K$, the resonant Rabi frequency; and the measured rate-based midpoint is κ_ret/K = 1.3–2.0 across two decades of Γt. The location claim holds, with K read as the population-transfer rate, to within a factor 1.6. Two qualifications the test adds. The location is a statement about *rates*: read from recoverability at a fixed observation time it drifts outward, from 1.7 to 5 as Γt goes from 0.15 to 10, because exp(−2Γ_eff t) reaches a given value at ever smaller occupation. And the crossover's relative width in κ_ret/K is of order one, 25–75 % points at 1.2–3.

3.3 The width, and what it is a width of

The layer is the set of deficits within which the site's restoring rate is below the coupling, $\Delta E \lesssim \hbar K$, i.e. a *share* width $\mathbf{w = K/\omega}$. In the deficit-rate variable the same layer is the crossover of relative width order one that the test measured; the two are one statement in two variables, and the carrier frequency ω enters only through the conversion between them. That was checked directly: removing the rotating-wave approximation from the absorber (the quantum Rabi model, photon Fock space to $n \leq 3$, the same record channel) lets ω into the dynamics, and it enters exactly as the Bloch–Siegert displacement of the resonance, −(1.1–1.5) K^2/ω, while the half-width of the crossover stays at 1.6 K to within 2 % from ω/K = 64 down to 8. The carrier is a shift, not a width.

Under the passive-absorber proxy K ∼ Γ (coherent coupling of order the measured linewidth, defensible where the coupling to the field is the interaction that sets the linewidth, and dropped where K is engineered independently), the layer width is tabulable:

System	Γ (s^{-1})	ω (s^{-1})	w	character
Alkali D-line, natural width [Steck 2025]	3–6 × 10^7	2–3 × 10^{15}	∼ 10^{-8}	razor-sharp

System	Γ (s^{-1})	ω (s^{-1})	w	character
Optical clock transition (^{87}Sr, Al^{+}) [Ludlow et al. 2015]	10^{-3}–10^{-2}	3–7 × 10^{15}	10^{-18}	sharpest in any laboratory
Transmon, $1/T_2$ [Kjaergaard et al. 2020]	10^{4}–10^{5}	3 × 10^{10}	10^{-7}–10^{-6}	atomic-grade, in a macroscopic circuit
Quantum dot, 4 K [Kuhlmann et al. 2015]	10^{9}–10^{11}	2 × 10^{15}	10^{-6}–10^{-5}	intermediate; the natural dial
Si photodiode / SPAD, 300 K [Sabbah & Riffe 2002]	10^{13}–10^{14}	3–5 × 10^{15}	10^{-3}–10^{-2}	percent-level layer — *v1.2: withdrawn from the candidate; a band absorber has no transition for a site to be detuned from, and its crossover is record-set ($\Gamma/2$)*

The table dissolves a familiar puzzle: superconducting circuits are macroscopic by particle count yet violate Bell and Leggett–Garg inequalities as clean two-level systems, because their w is atomic-grade. Mass and particle number are not the variable; κ_ret/K is.

3.4 Three stages, one crossing

stage	dynamics	recoverability
capture	unitary, distributed excitation across candidate sites in proportion to local intensity	recoverable — demonstrated: spin echoes, photon echoes, atomic-frequency-comb memories [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009], and the catch-and-reverse of a monitored quantum jump [Minev et al. 2019]
commitment	the absorption vertex and thermalization at one site: in a semiconductor the interband vertex, 10 fs–1 ps	**in-principle recoverability ends here**
registration	reservoir-powered amplification: avalanche, hotspot, readout	irrecoverable and now macroscopic; multiplies the vertex weight by a stochastic, position- and energy-dependent efficiency and adds photon-independent clicks (v1.2, per the SNSPD and SPAD literature); carries no interference-sensitive weight

Commitment is the vertex, not the record. Everything downstream is a deterministic cascade that copies what the vertex decided and contributes no statistics of its own; the commonplace that collapse happens when the record is amplified misassigns the cut by one stage. The placement was forced by an audit of detector timescales: for silicon the commitment window is marginal against any selection dynamics and for superconducting nanowire absorbers it is inverted, so no downstream clock can move the outcome weights, and the SPAD/SNSPD asymmetry a gap-gated reading once predicted does not arise. Cooling widens the reversibility window without moving the crossing only in the coupling-set regime $\Gamma \ll K$ (§6; v1.2).

Two physical requirements the tests added. **A dense record channel.** In the exact model one record degree of freedom coupled to the site gives coherent exchange — the return oscillates in the coupling strength and never settles — four modes leak nothing, sixteen recur within the capture time, and only a channel dense on the capture timescale (recurrence time $2\pi N/B \gg t$) behaves as the golden-rule continuum in which what leaks does not return. A "record" is a cut only where it is a continuum on the capture timescale. **The amplifier's engaged regime leaves recoverability untouched.** In the injected self-sustaining oscillator that models an amplifier, swept through its gain, the stored energy — what a record channel leaks — stays at the field-set value F^2/Δ^2 from the linear regime through the point where a free phase first exists and through the point where the forced fixed point loses stability, with slope 0.004 per unit gain, and rises with slope 1.1 only past the onset of free running at $g = 0.355$, tracking the oscillator's own limit-cycle energy. The handover of energy bookkeeping from field to site is continuous, with a corner and no jump; only the phase-winding observable switches. A self-sustaining absorber kinks the recoverability crossover; it does not sharpen it. This is amplifier phenomenology, retained as such; it is no longer the mechanism of anything.

3.5 What recoverability is: the reference trichotomy, and the one claim that is this paper's own

A reversal requires the stored phase history to point back along the path; a wavefront can be run backward only by an agent holding its complete phase record, as phase-conjugate readout demonstrates. Whether the record survives is set by the *reference structure* of the medium:

reference structure across sites	record	consequence
common (hologram, homodyne local oscillator, comb)	invertible	reversible
different but deterministically known (inhomogeneous ensemble)	invertible by rephasing	echo-recoverable: capture without commitment
random and unrecorded (a thermal detector bulk; each exchange hop)	non-invertible	history diffuses; commitment possible

The middle row is why echoes work; the $T_2{}^*/T_2$ split of magnetic resonance is this trichotomy under its textbook name. The bottom row is made by heat: thermal motion delivers phase kicks at random, unrecorded times, and the record of the incident wave's history spreads over e^{S/k_B} configurations of the medium, S growing at the dephasing rate. A 500 nm photon thermalized at 300 K produces about 96 k_B of entropy and spreads its record over some 10^{42} microstates within a picosecond; Landauer's bound then prices the reversal at $T\,\Delta S = E$, the photon's own energy [Landauer 1961]. Pastawski's polarization echoes exhibit the strong form, in which past a point the decay is set by the medium's own many-body dynamics and a better reversal does not help [Levstein et al. 1998; Pastawski et al. 2000].

Here is the paper's one claim that decoherence theory does not already make, and it is stated as an argument, not a result. Decoherence says the interference terms survive in the global state and are inaccessible *for all practical purposes*; Bell would not accept the phrase. This paper says that when the references are random and unrecorded, no agent — not merely no practical agent — holds what a reversal requires, and recoverability ends *in principle* at the vertex. The argument is the trichotomy's bottom row plus the observation that the reversal would cost the deposited energy back. Its weakness is plain: 10^{42} is astronomical, not infinite, and a Poincaré recurrence is not forbidden. We hold the claim as the paper's, flag it, and note that everything else in this section stands without it.

v1.1: conceded. Three reviewers showed independently that the claim does not survive: the environment is excluded from the reversal by the criterion's own definition, so the in-principle ending is definitional; Landauer's bound prices erasure, not reversal, which carries no such minimum for an agent holding the microstate [Bennett 1973, 1982]; the polarization-echo results concern how imperfect reversals fail; and a Poincaré recurrence is not forbidden. Recoverability of the reduced dynamics ends for all practical purposes, as decoherence says; what ends it in principle is the event of §4.1, which is postulated, and nothing else. Open problem 2 is withdrawn.

3.6 The detector taxonomy

Two numbers span the detector zoo: the per-event capture strength and the commit threshold k, how many captures must accumulate at one site within its memory time before commitment. The photon-counting detector is the projective limit, k = 1: one quantum, one vertex, one click, the only class for which per-event Born claims are properly made. NMR/MRI is the ensemble limit, no single-quantum threshold, every free-induction decay a watched relaxation. The cloud chamber is the repeated limit, not one drawn-out measurement but a sequence of capture–commit–register triplets (Mott's problem [Mott 1929], resolved as track-recording taxonomy). The photographic emulsion is the integrating limit, k ≈ 3–4 [Gurney & Mott 1938], and its reciprocity failure — a sub-threshold latent speck decaying if not reinforced — is everyday evidence that commitment and registration are physically distinct: progress toward a record can be made and then lost. The audit that makes the anatomy calorimetric: every click deposits the full quantum at one site, at every fringe position; the distributed capture is reversible energy, returned at κ_ret, never banked; losing sites retain no residue. The budget closes at one address.

4. Selection at absorption: the postulate and its price

4.1 The postulate, in the language of a sink

(The sponsor's formulation, 2026-09-04; the formal statement follows it and says the same thing.) A quantum arrives at a detector as a wave spread over many sites. Three things then happen, and the paper postulates all three.

1. **Every site draws, in proportion to the square.** A detector site at the single-photon level is a linear absorber: it has no phase of its own, the wave creates its oscillation, and the phase difference between the two is fixed by the site's detuning and return rate rather than settled by any dynamics. The energy such a site draws from the field is proportional to the square of the local amplitude, weighted by a Lorentzian in its detuning whose width is the coupling (the occupation $2K^2/(\Delta^2 + 4K^2)$ of §3.2). Eligibility is therefore graded, not sharp: no site is excluded, and a far-detuned site simply draws little. The sharp threshold of a self-sustained clock's tongue — eligible inside, nothing outside — is what a linear absorber does not have, and §4.2 says what that sharp threshold costs.

2. **A sink opens at one of them, by chance.** At each eligible site a sink can open at any moment, with a probability per unit time proportional to the energy that site has drawn so far, the same rule at every site. The first sink to open anywhere in the eligible set is the one that opens. Two phrases are load-bearing. *By chance*: the sink does not open when the site's phase reaches alignment with the wave. That rule makes the winner the fastest of N nearly deterministic slides, whose statistics are wrong (§4.2: exponent 1.5, and the count of eligible sites enters only as its logarithm). The sink opens as a memoryless event whose rate the absorbed energy sets, and then the count enters in proportion. *Drawn so far*: the rate is linear in that energy, hence proportional to the squared amplitude at the site. A square-root rate or a quadratic one gives the wrong statistics too, and the rate's timing — whether it lags the absorption — does not matter.

3. **The whole quantum flows through it.** Once a sink is open, the quantum's entire energy passes through that one site, and the amplitude at every other site vanishes — including sites in the other arm of an interferometer, metres away — within a thousandth of the time the sink took to open, faster than light crosses the detector. One sink per quantum. This is the one-quantum constraint. The sink picture names it; nothing derives it.

Stated formally. At the vertex, commitment is a stochastic event whose hazard at a site is linear in the energy the site has absorbed from the field, hence proportional to the squared amplitude there, with a time profile that does not depend on the site; exactly one site completes; and the constraint that enforces "exactly one" acts across the detector faster than light crosses it. Clause 1 is the linear response, the golden rule's matrix element squared; clause 2 is Fermi's golden rule [Dirac 1927; Fermi 1950] read as an event rather than as a rate, and it is what makes the outcome frequencies Born; clause 3 is the nonlocality that one world costs (§5). The paper derives none of the three. It took the program two months and a ledger of simulations to establish that it could not, and the postulate is written in the form the simulations said any mechanism would have to supply.

v1.1: clause 2 corrected. A hazard on the whole quantum's survival that is linear in the energy a site has drawn — the wording above — is falsified by the record's own race as soon as the two channels are reached at different times (`heisenberg_cut_recoverability/REVIEW_RUNS_RESULTS.md`, Run B): with channel B's pulse delayed past channel A's, A wins in every trial in which it fires (1.000 at splits from 10° to 45°, 0.994 at 60°, 0.912 at 70°) where Born gives it 0.97 down to 0.12, and a channel's click probability is an exponential of its intensity rather than linear (a 36 % excess at half intensity). The rule that reproduces quantum mechanics is the one in which the sink opens at the golden-rule rate times the site's *branch weight in the conditional state* — the state renormalized by every null elsewhere — which is the quantum-jump unravelling of photodetection [Srinivas & Davies 1981; Dalibard et al. 1992; Carmichael 1993; Wiseman & Milburn 2010] read as what happens. That is the Born rule's branch weights put in, and the null-conditioned renormalization is itself nonlocal. The short form carries the restated postulate and its stochastic master equation; this paragraph records the correction against the wording above.

4.2 What was learned by trying to derive it

The mechanism tested was a race among noisy Adler clocks, each detector site a phase oscillator pulled toward the field by a coupling proportional to the local amplitude, commitment going to the first clock to lock. With the plan's own amplitude-neutral commitment rule (a fixed dwell inside the tongue), the race gave a coupling exponent of **1.56** [1.44, 1.69] against Born's 2 — and the reason was computed: each clock's commitment is a near-deterministic slide from its random starting phase, and the fastest of N such slides gains only logarithmically in N where a Poisson race gains a full power; the deterministic skeleton with no noise reproduces the exponent. The exponent is criterion-dependent (1.4 to 1.8 across dwell, tolerance, noise, and pulse), tracks the detuning spectrum as tongue width times the square root of the rate, overshoots to 3.8 with a dwell inversely proportional to the coupling, and reaches 2.00 only with a dwell tuned as the inverse quarter power of the coupling — a fitted criterion, not a mechanism. The Born curve appears, **2.16** [2.14, 2.19] within 0.02 of Born at every angle, exactly when commitment is made memoryless with a hazard linear in absorbed energy under the clock's own power law and no inserted square: the square then arises as tongue width ($\propto$ K) times locked absorption rate ($\propto$ K). But memorylessness and hazard linearity are the golden rule's structure, put in by hand; the hazard's *form* is load-bearing (a square-root hazard gives 1.70, a quadratic one 3.25) while its *timing* is not (a decade of hazard memory changes nothing). And exclusivity is not derived and cannot be local: independent site rates give a coincidence ratio of 1 at a balanced split against the observed 2×10^{-3}, and the one-quantum stop reproduces the data only if it acts within about 10^{-3} of the commitment time, orders of magnitude faster than light crosses the port separation.

Net: a substrate of phase-locked clocks can realize the Born measure only by containing a commitment process with the golden rule's structure and a nonlocal one-quantum constraint, neither of which it produces. What the substrate supplied on its own was a geometric origin for a square — width times rate — which is a real observation about where a square *can* come from and not a derivation of the rule. The positive residue is the specification in §4.1, each clause established with a number, which is handed to whoever tries next.

Where the substrate's square comes from, and what that costs (added the same day, after the sponsor asked whether synchronization becomes Born-compatible once deriving Born is no longer required). It does, on one condition that turns out to be a fine-tuning. The gated race's second power of K is the tongue factor, the fraction of clocks whose detuning lies inside the coupling; narrow the substrate's detuning spread and it disappears. Run on Gaussian spectra of decreasing width with the same golden-rule hazard, the exponent falls monotonically from 2.21 on the flat grid through 2.09, 2.01, 1.78, and 1.52 at widths 2, 1, 0.5, and 0.25

(in units where the couplings run to 2) to 1.08 with every clock at the same detuning, and the energy a channel absorbs tracks the exponent at every width. It equals 2 at exactly one width, half the maximum coupling, where the Born comparator is accepted; on the flat grid it is rejected, a 2.7 % excess at the 30° split, outside the precision of Malus-law tests by orders of magnitude. So synchronization plus the golden-rule hazard is Born-*compatible* in the sense that some spectrum makes it so, and not in the sense that matters: a universal Born rule would need every substrate at the same tuned width. And the narrow-band limit is a prediction that differs from quantum mechanics — two identical absorbers each strongly coupled to one arm of a split photon, coupling exceeding any spread, would divide the photon in proportion to the amplitude rather than its square — in a regime, single-photon strong coupling with identical atoms, where nothing in the record suggests a linear law. That the failure has not been seen in ordinary detectors is because a single photon's coupling is small against any inhomogeneous width, which is the broadband case. The broadband limit was then measured directly, with the predictions on record that its excess above 2 was a transient of grid, window, or pulse and would go away: it did not. On 64 clocks, with stationary windows from 4 to 32, pulses from 1 to 16, noise on or off, and commitment pushed from 0.6 to 14 time units past the locking transient, the exponent sat at 2.13–2.16 in every configuration, while the channel's absorbed energy did fall toward the semicircle's K^2 as the window lengthened. The identity that makes the broadband limit Born-exact governs the energy, not the outcome. Three explanations of the residual failed in one evening; a fourth run, with the noise off at the longest commitment time, brought the exponent to 2.09 with the absorbed energy at K^2.03 and split the residual into a small part the noise owns and a larger deterministic part that decays only logarithmically with the commitment time — the signature an extreme-value ingredient, sites that begin near their locked phase drawing first, would leave. Its mechanism is an open item of the record, not of this paper. What the linear absorber makes plain is where the residual lives: with a sharp eligibility threshold the second power of K is supplied by the tongue's width, and everything that blurs the threshold — the pulse, the locking delay, the grid — moves the exponent; with the graded Lorentzian weight of §4.1 clause 1 there is no threshold to blur, the square is per site, the site count is independent of K, and the exponent is 2 exactly. Synchronization was never needed for the square; it was one way to manufacture it, with an error of a tenth of a power attached. This paper takes quantum mechanics' side of the discriminator, and it is the second reason, after the negative result, that synchronization is a model class here and not the physics of the vertex.

The peak's width in physical units, and the atoms under it (added at the sponsor's request; order-of-magnitude, with the assumptions stated). The peak above the absorption threshold is a width in detuning, not in space: a site is eligible when $|\Delta| < K$, so the peak is $2K$ wide and its area, $\pi K^2/2$, is a summed absorption rate. K is the single-photon Rabi frequency at the site, $K = d\,E_1/\hbar$ with $E_1 = \sqrt{\hbar\omega/2\epsilon_0 V}$ the field of one photon in its mode volume V, the spot area times the pulse length. A nanosecond photon at 500 nm focused to a one-micron spot has $V \approx 3 \times 10^{-13}$ m^3 and $E_1 \approx 270$ V/m; a strong atomic transition dipole of two electron-ångströms then gives $K \approx 7 \times 10^7$ s^{-1}, a peak about 20 MHz wide. A hundred-femtosecond photon is 10^4 times more concentrated and gives a peak near 2 GHz. Against these, inhomogeneous spreads run from gigahertz in a vapour to terahertz in solids. The spatial circle is the photon's spot, and the atoms it covers depend on the medium; those inside the peak are that count times the fraction of the spread the peak occupies:

medium (one-micron spot)	atoms in the spot	spread (s^{-1})	fraction inside the peak	eligible atoms
rubidium vapour, 1 cm path	10^2–10^3	Doppler, $\sim 3 \times 10^9$	$\sim 5\,\%$	tens or fewer
rare-earth-doped crystal, mm absorption depth	$\sim 10^{10}$ ions	$\sim 3 \times 10^{10}$	$\sim 5 \times 10^{-3}$	$\sim 10^7$
silicon, 1 µm absorption depth	$\sim 5 \times 10^{10}$	band continuum	undefined	all, in the only sense available

The last row is the one that matters for detectors. A band absorber has no single transition frequency for a site to be detuned from — any site can absorb any energy above the gap — so the peak's width has no meaning there, every site is eligible, and the substrate is broadband in the fullest sense: exactly the limit in which the race's exponent sits above 2 and not at it.

4.3 The nearest neighbour: the transactional interpretation

Selection at the interaction, from the interference of wavepackets, is the transactional interpretation, and the relation is drawn here exactly because the sponsor's original intuition — actualization at the absorber, from interference — lands there. Cramer's interpretation [Cramer 1986], built on Wheeler–Feynman absorber electrodynamics [Wheeler & Feynman 1945], describes a quantum event as a transaction between an emitter's retarded offer wave ψ and an absorber's advanced confirmation wave ψ*, and reads the Born weight as their product. Cramer notes that the relativistic wave equations, Dirac's and Klein–Gordon's, have advanced as well as retarded solutions and that the Schrödinger equation is their non-relativistic limit, so that the negative-energy branch is the confirmation wave's home and* $\psi \to \psi$ is time reversal; he gives the absorber no microscopic description, treating it functionally as what precipitates the collapse. Kastner's relativistic version [Kastner 2012] adopts Davies' direct-action quantum electrodynamics [Davies 1971, 1972], in which the field is replaced by direct current-to-current interaction between fermionic (Dirac) currents; it treats the coupling amplitude as what governs the generation of offers and confirmations and, by its own statement, ignores the spin of the fermions and defers a full treatment with spinors. Kastner and Cramer [2018] then define an absorber as an elementary bound system, an atom or molecule, derive the Born rule for radiative processes as the product of offer and confirmation amplitudes over N ground-state atoms, and state that absorption may be precisely quantified for atomic electrons in ordinary QED; the absorber's definition has been criticized as insufficiently specified [Boisvert & Marchildon 2013]. So: **no one in the transactional lineage has included the Dirac spinor structure in a microscopic description of the absorber.** Cramer used the Dirac equation's advanced solutions; Kastner used Dirac currents as sources and set spin aside; the absorber is an atom with a QED coupling, never a detector with a record channel.

What is shared with this paper: actualization at the absorber; one world; nonlocality; and a square that comes from an interference-like product rather than from a postulate about probabilities. What differs: this paper has no advanced waves; it locates the cut, gives it a width, and has tested the location; it places commitment at the vertex with the dense-record requirement and the timescale audit; and it does not present the square as derived. On that last point the two are more alike than the transactional literature says: $\psi\psi^*$ is a rule about products, as the golden-rule hazard is a rule about rates, and in both the square is supplied by the structure of the rule. The transactional interpretation has never produced a prediction that differs from quantum mechanics. Neither has this one, and §7 says so.

4.4 The price, itemized

The postulate costs three things. **Nonlocality**, in Bell's sense and never signalling: the one-quantum constraint acts across a detector faster than light. **A stochastic element** that is not in the wave equation: the event is not unitary, and the paper does not say what makes it happen — only where, when, and with what hazard. **A preferred foliation** to order nonlocal commitments (§5). Against this it buys one world, a located cut, and checkability: the premises are about detectors, which laboratories can interrogate, rather than about worlds, which they cannot.

5. The entangled sector and the frame

(Conditional on the real-wave commitment extended to the non-separable two-particle configuration, over and above the single-detector commitments; the configuration-space objection lives here and is held open. The reviewed drafts carried a picture of the nonlocal update, a rotation of a shared frame; it is withdrawn at the sponsor's decision of 2026-09-04, and this section now states only what the theorems require.)

Two objects, not to be blurred. The local thermal reference at each detector — the random-unrecorded references that make commitment possible — is established in the common past of both wings and is independent of the analyzer settings: it is a Bell common-cause variable, and by Bell's theorem it cannot source the correlations; a clock-as-local-variable model of the outcomes is sub-classical, CHSH $\leq \sqrt{2}$. The correlation comes from the second object: the extended, non-separable field configuration carrying the entangled two-particle mode across both wings — not a common cause but one configuration of the real field spanning A and B. Locating the nonlocality there adds no postulate; it restates the real-wave commitment. The framework is not superdeterministic (settings are free), and no-signalling holds because the ensemble average of the postulated jump process is a Lindblad equation, linear in the density matrix, which is the condition under which a stochastic reduction does not signal [Gisin 1989; Polchinski 1991] (v1.1: the v1.0 wording — that it "follows from the hazard's linearity in the squared amplitude" as Gisin's 1990 argument in reverse — was wrong; the relevant condition is linearity of the ensemble map, and a hazard linear in local energy would signal, as the reviewers' Jensen-inequality counterexample shows; the corrected postulate's average is Lindblad by construction).

The foliation, priced. Nonlocal commitments must be ordered, and the ordering is a preferred slicing of the substrate that enters through the measurement sector only: capture is exactly Lorentz covariant, and the frame appears where the event does. As written this is explicit low-energy Lorentz violation confined to measurement. Ordinary matter couples to it, beyond the electromagnetic commitment itself, only gravitationally — the suppression that explains why clock-comparison and Michelson–Morley bounds have not registered it — but the quantitative demonstration that the residual anisotropy clears every established bound remains owed, alongside the covariant formulation of the selection event. Whether the frame is a vacuum rest frame or a matter rest frame is a fork the record holds open (ledger E-12). *v1.1:* the slicing is a choice with alternatives, not a necessity — Tumulka's flash-ontology GRW, Bedingham's and Pearle's relativistic CSL, and the state-derived foliation of Dürr and colleagues order, or avoid ordering, nonlocal collapses covariantly [Tumulka 2006; Bedingham 2011; Pearle 2015; Dürr et al. 2014] — and by the paper's own no-signalling it is unobservable in any correlation, exactly as Bohm's is; the v1.0 sentence that §7 "is built to ask" is withdrawn.

6. Time, energy, information

(Interpretive; sponsor-originated.) Three quantities are bound and not identified. A phase's *rate* is an energy, $E = \hbar\, d\varphi/dt$; its *position* is the informational variable, the thing a record can capture and the thing thermal matter leaves random; the phase itself is time. Three bounds make the binding quantitative. Time–energy uncertainty: a phase cannot be read to Δt without an energy spread $\hbar/\Delta t$. Margolus–Levitin: the energy sets the maximum rate, $2E/\pi\hbar$, at which a system passes between distinguishable states — energy is the clock rate of information. Landauer: erasing one position costs k_BT ln 2. So ħ and k_BT are the two exchange rates and there is no third; one quantum can carry unboundedly many bits and one bit can ride on arbitrarily little energy, which is why the three are bound and not one. The three come apart at exactly one place: energy is conserved through registration while position information is not. That is the cut, stated in information language, and §3.5's Landauer price is the same statement with numbers: the reversal of a thermalized quantum costs the quantum back.

Temperature as a dial on the record channel. Temperature is a proxy; the physical variable is the rate $\Gamma(T)$ of unrecorded phase kicks, rising with T through phonon occupation in a solid, as $n\sqrt{T}$ through collisions in a gas, and falling as T^{−3/2} through Coulomb collisions in a plasma. Existing temperature studies fix the qualitative content: rare-earth optical coherence collapsing from a millisecond at 2 K by 10 K under known phonon laws [Equall et al. 1995; Könz et al. 2003]; fullerene fringes fading as the molecules' internal temperature rises and they emit photons that record their path [Hackermüller et al. 2004]; transmon coherence falling through activated quasiparticles. The exact model gives $R = \exp[-2\Gamma\ t\ f(\Delta)]$ with f the site's occupation, independent of the channel's bandwidth at fixed Γ to one part in 10^3, and therefore two separable predictions: retrieval efficiency versus storage time at several temperatures collapses onto one curve in $\Gamma(T)\cdot t$ if the channel is memoryless (a non-collapse measures non-Markovian bath structure); and the retrieval crossover in input detuning sits at $|\Delta| \approx 2K$ and does not move with temperature while $\Gamma(T) \ll K$, drifting inward by at most a factor 1.5 as Γ approaches K and never outward. Both coincide with Markovian decoherence theory; the test cannot discriminate this interpretation, but the interpretation owes the prediction, and this is it: the crossing is coupling-set only while the coupling exceeds the record channel's rate (v1.1, Run A: for $\Gamma \gg K$, the regime of every room-temperature semiconductor detector, the crossover sits at $\Gamma/2$ and is record-set and hence temperature-set); in the coupling-set regime cooling widens the window without moving it.

7. The experimental program, and what falsifies what

Everything here is either a consistency test that quantum mechanics also passes or a test of one of this paper's own premises; the paper claims no prediction that differs from quantum mechanics.

The coupling dial. The cut is placed on κ_ret/K, not on mass, size, or gravitational self-energy; the collapse programs (CSL, Diósi–Penrose) place their boundary on mass. An engineered system moved along the coupling dial at fixed mass discriminates the two: the preregistrable protocol is a transmon coupled to a lossy readout mode, where the two rates are independently engineered, sweeping the coupling ratio and recording commitment latencies and the echo-recoverable window against a circuit-QED null model. The null model is computable today; the framework curve is now the exact model's crossover of §3.2, which predicts smooth variation co-located at one ratio — and, being decoherence-compatible, predicts the same as the null. Honestly stated: this dial discriminates the interpretation from the mass-based collapse programs, not from decoherence. Macromolecule interferometry, the collapse programs' home turf, is re-read accordingly: interference should be lost where the coupling budget says so, and the fullerene thermal-emission result already reads that way.

The last deviation channel. Any local POVM — quantum mechanics with arbitrary physics inside each port — makes the conditional statistics of a photon split S:(1−S) into two ports of mismatched structure exactly affine in S. Curvature κ S(1−S) would overturn the linear structure of quantum mechanics itself; a null at the 10^{-3} level closes the channel. Our own judgement favours the null; the bench decides, and the channel is listed because a program that once carried it owes the reader its fate.

The energy audit. Full quantum per click at every fringe position, no calorimetric residue at losing sites — a consistency test energy-resolving detectors (transition-edge sensors, kinetic-inductance cameras) could begin now, framed by no experiment yet.

The temperature collapse test of §6, on a rare-earth photon-echo memory at variable input detuning and three or four temperatures.

The frame's tests. Because the foliation enters through measurement only, its tests are boundary tests, in the dissipation channel that existing free-precession bounds (neutron bounds at 10^{-34} GeV) do not probe:

	test	probes
T1	gravitational Bell vs. linewidth	the H' postulate
T2	sidereal GHZ	cosmic vs. local frame
T3	AB visibility vs. dissipation	in-flight frame coupling
T4	MRI/NMR surface relaxation	frame coupling in the dissipation channel
T5	LIGO sidereal quantum-noise modulation	cosmic frame, boundary anisotropy

The gravitational Bell candidate is kept with its honesty apparatus intact: two Bell detectors at different potentials with narrow-linewidth photons would show CHSH degrading as $\exp(-\delta\varphi^2/2)$ with $\delta\varphi = \omega\ \Delta\Phi/c^2\ \Delta\nu$ — *if* the local reference couples into the polarization projection through an additional, non-covariant term H'. The effect does not follow from the framework plus QED, where the redshift cancels as a common-mode phase; the experiment tests that postulate, not the core. The Bose–Marletto–Vedral configuration discriminates the wave-energy reading of the ontology (a real-energy-density source would not gravitationally entangle two masses) and is noted as far-future.

What falsifies what. Curvature in the port discriminator or an above-Tsirelson excess would overturn quantum mechanics' linear structure; their nulls close channels and leave the core intact, since the core is built to be Born-exact. A mass-located boundary on the coupling dial falsifies this paper; a coupling-located one falsifies the mass-based collapse programs. *v1.1: the frame tests are withdrawn as tests of the interpretation.* Since it does not signal, the foliation modulates no correlation and a sidereal null retires nothing; the before-before and 24-hour experiments exist and are null [Zbinden et al. 2001; Stefanov et al. 2002; Salart et al. 2008]. The H' candidate is a separate, non-covariant postulate, and its test is a test of that postulate only. A linear absorber with a dense record channel whose recoverability crossover sat far from κ_ret/K = 1 would have falsified the location in the simplest model; it sat at 1.3–2.0.

8. Placement

account	one outcome?	mechanism	new constants	price
Decoherence-only	no (redescribes)	none needed	no	the event is not explained; the cut is FAPP
Everett	no (branching)	none	no	many worlds; Born weights via decision theory, contested
GRW / CSL / Diósi–Penrose	yes	postulated noise / gravity	yes	new physics, now largely excluded by X-ray and underground bounds [Bassi et al. 2013; Donadi et al. 2021]
Bohm / Nelson	yes	guidance law / stochastic mechanics	no	hidden configurations; inaccessible foliation
Transactional (Cramer, Kastner)	yes	offer–confirmation transaction	no	advanced waves; the square as a product rule
ABN (Curie–Weiss)	registration only	statistical mechanics	no	selection assumed at the crucial step [Allahverdyan et al. 2013]
This paper	yes	postulated event at the vertex, golden-rule hazard	no	wave realism; a nonlocal one-quantum constraint; a foliation; no derivation of the square

The family is the single-world, ψ-ontic, nonlocal one; Bohm, Nelson, and the transactional interpretation are its neighbours. The distinction this paper draws within the family is *where the measure sits*: not over hidden system configurations (Bohm) and not in a product of forward and backward waves (Cramer), but in a hazard at a detector site with a physical address, a tested location, and a computable width — a measure about detectors, which laboratories can interrogate. Against Penrose and Diósi the operational split is clean: they put the boundary on mass, this paper on coupling, and the dial decides. Against Montevideo and the relational-clock tradition, the closest kin in spirit: their coherence loss is fundamental, ours is environmental and dissipative, absent for a closed system however imprecise its clock. Against decoherence-only, the paper accepts every einselection result wholesale, adds the one thing that programme's own reviews concede is missing, and concedes in turn that its addition is a postulate.

9. Open problems

1. **The quantum self-sustaining absorber with a record channel.** The exact tests used a linear absorber (§3.2–3.3) and, for the amplifier, a classical oscillator with gain put in by hand (§3.4). The one calculation that would show the three regimes — slaved, engaged, running — with κ_ret a coherent return rather than a damping is an open quantum system with a limit cycle and a dense record channel; the quantum synchronization literature has the oscillator and not the channel.
2. **In principle versus for all practical purposes.** *Withdrawn in v1.1:* conceded to decoherence (§3.5).
3. **The covariant formulation.** The selection event on the foliation, with its no-signalling proof written independently of the statistics it reproduces, and the quantitative clearance of every established Lorentz bound (§5).
4. **The configuration-space objection.** Schrödinger's second defeat, held open at the non-separable two-particle configuration: a premise priced, not a problem solved.
5. **The port discriminator's fork** (§7): whether a deeper principle closes the last deviation channel or the curvature is real; the bench decides.
6. **The location in a self-sustaining sub-threshold system.** In the forced oscillator below threshold, K is set by the response itself, so κ_ret/K = 1 cannot be tested there — the circularity the second paper flagged — and the exact test resolved it only for a linear absorber.

Retired from the ledger: the derivation of the Born weights from substrate dynamics (recorded as a negative result), and the continuous boundary-spanning model exhibiting a *bifurcation* at κ_ret/K = 1 (the boundary-spanning model exists and exhibits a crossover; the bifurcation was not there to find).

10. What this is

Schrödinger built an equation that describes the evolution of the wave and meant it as physics. The description was extended to relativity and to fields, and then stopped, one postulate short, for a century. This program set out to show that the stopping was contingent — that the measurement event was the same substrate doing one more thing waves do — and found that it could not: the last postulate did not become dynamics under the mechanism proposed, and the record of that finding is published on purpose. What survives is an interpretation with an address. The wave is real and it evolves; where it is absorbed, one site commits, in one world, at a place and a time the physics of the detector fixes; the cut is there, with a width the coupling sets; and what the postulate does not explain is stated as a postulate, priced, and handed forward with the specification any successor must meet.

Appendix A — The record

- `NEGATIVE_RESULT.md` — what was attempted, what failed, what was learned, what remains.
- `drafts/PAPER1_DRAFT_born_selection.md` (v0.9) — the selection game and its claim boundary as it stood when the program ended; `drafts/EQUATIONS_RECONCILIATION_LEDGER_2026-09-01.md` — every finding, uncertainty first, including the wrong predictions.
- `adler_two_channel_exploratory/RESULTS.md` — the race, its sensitivities, its Born-producing variant, and its non-claims; `validation/` — the numerical campaigns, overrides, re-freeze and re-decision, each a hashed record.
- `drafts/PAPER2_DRAFT_heisenberg_cut.md` (v0.4) — the cut paper as absorbed here, with its seven marked changes; `heisenberg_cut_recoverability/` — the exact tests, predictions first, results scored.
- `drafts/PAPER3_DRAFT_dk_framework.md` (v0.1) and `current_revision_DK_paper.md` (v8) — the synchronization framework and its predecessor, frozen as the provenance record.
- `GLOSSARY.md` — the vocabulary in plain terms.

References

- Afzelius, M., Simon, C., de Riedmatten, H., & Gisin, N. (2009). Multimode quantum memory based on atomic frequency combs. *Phys. Rev. A* **79**, 052329.
- Allahverdyan, A. E., Balian, R., & Nieuwenhuizen, T. M. (2013). Understanding quantum measurement from the solution of dynamical models. *Phys. Rep.* **525**, 1–166.
- Bassi, A., Lochan, K., Satin, S., Singh, T. P., & Ulbricht, H. (2013). Models of wave-function collapse, underlying theories, and experimental tests. *Rev. Mod. Phys.* **85**, 471–527.
- Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika* **1**, 195–200.
- Bell, J. S. (1990). Against "measurement". *Physics World* **3**(8), 33–40.
- Bialynicki-Birula, I. (1996). Photon wave function. *Progress in Optics* **36**, 245–294.
- Boisvert, J.-S., & Marchildon, L. (2013). Absorbers in the transactional interpretation of quantum mechanics. *Found. Phys.* **43**, 294–309.
- Brune, M., Hagley, E., Dreyer, J., Maître, X., Maali, A., Wunderlich, C., Raimond, J. M., & Haroche, S. (1996). Observing the progressive decoherence of the "meter" in a quantum measurement. *Phys. Rev. Lett.* **77**, 4887–4890.
- Camilleri, K., & Schlosshauer, M. (2015). Niels Bohr as philosopher of experiment: does decoherence theory challenge Bohr's doctrine of classical concepts? *Stud. Hist. Phil. Mod. Phys.* **49**, 73–83.
- Cramer, J. G. (1986). The transactional interpretation of quantum mechanics. *Rev. Mod. Phys.* **58**, 647–687.
- Davies, P. C. W. (1971). Extension of Wheeler–Feynman quantum theory to the relativistic domain I. Scattering processes. *J. Phys. A* **4**, 836–845.
- Davies, P. C. W. (1972). Extension of Wheeler–Feynman quantum theory to the relativistic domain II. Emission processes. *J. Phys. A* **5**, 1025–1036.
- de Riedmatten, H., Afzelius, M., Staudt, M. U., Simon, C., & Gisin, N. (2008). A solid-state light–matter interface at the single-photon level. *Nature* **456**, 773–777.
- Dirac, P. A. M. (1927). The quantum theory of the emission and absorption of radiation. *Proc. R. Soc. London A* **114**, 243–265.
- Dirac, P. A. M. (1928). The quantum theory of the electron. *Proc. R. Soc. London A* **117**, 610–624.
- Donadi, S., Piscicchia, K., Curceanu, C., Diósi, L., Laubenstein, M., & Bassi, A. (2021). Underground test of gravity-related wave function collapse. *Nature Phys.* **17**, 74–78.
- Equall, R. W., Cone, R. L., & Macfarlane, R. M. (1995). Homogeneous broadening and hyperfine structure of optical transitions in Pr^{3+}:Y_2SiO_5. *Phys. Rev. B* **52**, 3963–3969.
- Everett, H. (1957). "Relative state" formulation of quantum mechanics. *Rev. Mod. Phys.* **29**, 454–462.
- Fermi, E. (1950). *Nuclear Physics.* University of Chicago Press. (The "Golden Rule No. 2".)
- Foldy, L. L., & Wouthuysen, S. A. (1950). On the Dirac theory of spin ½ particles and its non-relativistic limit. *Phys. Rev.* **78**, 29–36.
- Gisin, N. (1990). Weinberg's non-linear quantum mechanics and supraluminal communications. *Phys. Lett. A* **143**, 1–2.
- Glauber, R. J. (1963). Coherent and incoherent states of the radiation field. *Phys. Rev.* **131**, 2766–2788.
- Grangier, P., Roger, G., & Aspect, A. (1986). Experimental evidence for a photon anticorrelation effect on a beam splitter. *Europhys. Lett.* **1**, 173–179.
- Gurney, R. W., & Mott, N. F. (1938). The theory of the photolysis of silver bromide and the photographic latent image. *Proc. R. Soc. London A* **164**, 151–167.
- Hackermüller, L., Hornberger, K., Brezger, B., Zeilinger, A., & Arndt, M. (2004). Decoherence of matter waves by thermal emission of radiation. *Nature* **427**, 711–714.
- Hahn, E. L. (1950). Spin echoes. *Phys. Rev.* **80**, 580–594.
- Haken, H. (1975). Cooperative phenomena in systems far from thermal equilibrium and in nonphysical systems. *Rev. Mod. Phys.* **47**, 67–121.
- Hanbury Brown, R., & Twiss, R. Q. (1956). Correlation between photons in two coherent beams of light. *Nature* **177**, 27–29.
- Joos, E., & Zeh, H. D. (1985). The emergence of classical properties through interaction with the environment. *Z. Phys. B* **59**, 223–243.

- Kastner, R. E. (2012). The possibilist transactional interpretation and relativity. *Found. Phys.* **42**, 1094–1113.
- Kastner, R. E., & Cramer, J. G. (2018). Quantifying absorption in the transactional interpretation. *Int. J. Quantum Found.* **4**, 210–222.
- Kjaergaard, M., Schwartz, M. E., Braumüller, J., Krantz, P., Wang, J. I.-J., Gustavsson, S., & Oliver, W. D. (2020). Superconducting qubits: current state of play. *Annu. Rev. Condens. Matter Phys.* **11**, 369–395.
- Könz, F., Sun, Y., Thiel, C. W., Cone, R. L., Equall, R. W., Hutcheson, R. L., & Macfarlane, R. M. (2003). Temperature and concentration dependence of optical dephasing, spectral-hole lifetime, and anisotropic absorption in Eu^{3+}:Y_2SiO_5. *Phys. Rev. B* **68**, 085109.
- Kuhlmann, A. V., Prechtel, J. H., Houel, J., Ludwig, A., Reuter, D., Wieck, A. D., & Warburton, R. J. (2015). Transform-limited single photons from a single quantum dot. *Nature Commun.* **6**, 8204.
- Kurnit, N. A., Abella, I. D., & Hartmann, S. R. (1964). Observation of a photon echo. *Phys. Rev. Lett.* **13**, 567–568.
- Lan, S.-Y., Kuan, P.-C., Estey, B., English, D., Brown, J. M., Hohensee, M. A., & Müller, H. (2013). A clock directly linking time to a particle's mass. *Science* **339**, 554–557.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.* **5**, 183–191.
- Levstein, P. R., Usaj, G., & Pastawski, H. M. (1998). Attenuation of polarization echoes in nuclear magnetic resonance: a study of the emergence of dynamical irreversibility in many-body quantum systems. *J. Chem. Phys.* **108**, 2718–2724.
- Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E., & Schmidt, P. O. (2015). Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637–701.
- Minev, Z. K., et al. (2019). To catch and reverse a quantum jump mid-flight. *Nature* **570**, 200–204.
- Mott, N. F. (1929). The wave mechanics of α-ray tracks. *Proc. R. Soc. London A* **126**, 79–84.
- Ollivier, H., Poulin, D., & Zurek, W. H. (2004). Objective properties from subjective quantum states: environment as a witness. *Phys. Rev. Lett.* **93**, 220401.
- Pastawski, H. M., Levstein, P. R., Usaj, G., Raya, J., & Hirschinger, J. (2000). A nuclear magnetic resonance answer to the Boltzmann–Loschmidt controversy? *Physica A* **283**, 166–170.
- Penrose, R. (1996). On gravity's role in quantum state reduction. *Gen. Relativ. Gravit.* **28**, 581–600.
- Sabbah, A. J., & Riffe, D. M. (2002). Femtosecond pump-probe reflectivity study of silicon carrier dynamics. *Phys. Rev. B* **66**, 165217.
- Schlosshauer, M. (2004). Decoherence, the measurement problem, and interpretations of quantum mechanics. *Rev. Mod. Phys.* **76**, 1267–1305.
- Schlosshauer, M., & Camilleri, K. (2008). The quantum-to-classical transition: Bohr's doctrine of classical concepts, emergent classicality, and decoherence. arXiv:0804.1609.
- Schrödinger, E. (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsber. Preuss. Akad. Wiss., Phys.-Math. Kl.* 418–428.
- Steck, D. A. (2025). Rubidium 87 D Line Data (rev. 2.3.3). https://steck.us/alkalidata.
- Tegmark, M. (1993). Apparent wave function collapse caused by scattering. *Found. Phys. Lett.* **6**, 571–590.
- Tonomura, A., Endo, J., Matsuda, T., Kawasaki, T., & Ezawa, H. (1989). Demonstration of single-electron buildup of an interference pattern. *Am. J. Phys.* **57**, 117–120.
- Wheeler, J. A., & Feynman, R. P. (1945). Interaction with the absorber as the mechanism of radiation. *Rev. Mod. Phys.* **17**, 157–181.
- Zeh, H. D. (1970). On the interpretation of measurement in quantum theory. *Found. Phys.* **1**, 69–76.
- Zurek, W. H. (1981). Pointer basis of quantum apparatus: into what mixture does the wave packet collapse? *Phys. Rev. D* **24**, 1516–1525.
- Zurek, W. H. (1982). Environment-induced superselection rules. *Phys. Rev. D* **26**, 1862–1880.
- Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.* **75**, 715–775.
- Zurek, W. H. (2009). Quantum Darwinism. *Nature Phys.* **5**, 181–188.

Added in v1.1 (the short form carries the complete corrected list):

- Bedingham, D. J. (2011). Relativistic state reduction dynamics. *Found. Phys.* **41**, 686–704.
- Bennett, C. H. (1973). Logical reversibility of computation. *IBM J. Res. Dev.* **17**, 525–532.
- Bennett, C. H. (1982). The thermodynamics of computation — a review. *Int. J. Theor. Phys.* **21**, 905–940.
- Carmichael, H. J. (1993). *An Open Systems Approach to Quantum Optics*. Springer.
- Dalibard, J., Castin, Y., & Mølmer, K. (1992). Wave-function approach to dissipative processes in quantum optics. *Phys. Rev. Lett.* **68**, 580–583.
- Dürr, D., Goldstein, S., Norsen, T., Struyve, W., & Zanghì, N. (2014). Can Bohmian mechanics be made relativistic? *Proc. R. Soc. A* **470**, 20130699.
- Gisin, N. (1989). Stochastic quantum dynamics and relativity. *Helv. Phys. Acta* **62**, 363–371.
- Pearle, P. (2015). Relativistic dynamical collapse model. *Phys. Rev. D* **91**, 105012.
- Polchinski, J. (1991). Weinberg's nonlinear quantum mechanics and the Einstein–Podolsky–Rosen paradox. *Phys. Rev. Lett.* **66**, 397–400.
- Srinivas, M. D., & Davies, E. B. (1981). Photon counting probabilities in quantum optics. *Optica Acta* **28**, 981–996.
- Stefanov, A., Zbinden, H., Gisin, N., & Suarez, A. (2002). Quantum correlations with spacelike separated beam splitters in motion: experimental test of multisimultaneity. *Phys. Rev. Lett.* **88**, 120404.
- Tumulka, R. (2006). A relativistic version of the Ghirardi–Rimini–Weber model. *J. Stat. Phys.* **125**, 821–840.
- Wiseman, H. M., & Milburn, G. J. (2010). *Quantum Measurement and Control*. Cambridge University Press.
- Zbinden, H., Brendel, J., Gisin, N., & Tittel, W. (2001). Experimental test of nonlocal quantum correlation in relativistic configurations. *Phys. Rev. A* **63**, 022111.